

SN6200-LD Engineering Sample

Release Notes

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SN6200-LD Engineering Sample

This document provides information about the SN6200-LD engineering sample for Cumulus Linux 5.18.1.

Important:

- > The SN6200-LD engineering sample requires Cumulus Linux release 5.18.6810. For information about Cumulus Linux 5.18.1, refer to the [Cumulus Linux 5.18 User Guide](#).
- > Future upgrades from the SN6200-LD engineering sample will be binary upgrades using ONIE.

Supported Platforms

The SN6200-LD engineering sample supports the following platforms:

Model Number	Part Number	Description
SN6810-LD	920-9N64K-00LI-GC0	Spectrum-6 based 800G Ethernet switch with Cumulus Linux Authentication, Co-Packaged Optics, 128 MMC-12, Single QSFP112, 8-Cores CPU , Secure-boot, 48VDC Busbar, Liquid-Cooling, Rail-Kit.

SN6200-LD Supported Port Speeds

Speed [GbE]	Auto-Negotiation	Force			Cables/Modules	Cable Length [m]
		RS FEC	FC FEC	NO FEC		
200G PAM4: 800G 400G 200G	✓	✓	X	X	Optic	Up to 100
					Copper	1
100G (service port)	✓	✓	X	X	Optic	Up to 100

- > PAM4 is the system default.
- > When connecting NVIDIA-to-NVIDIA in PAM4, make sure Auto-negotiation is enabled.
- > 100 Gbps serdes lane speed is not supported

Package Content

NVIDIA has validated this release with the following component versions:

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Package	Version
SDK	4.10.0722-017
Kernel	6.1.123-1
MFT	4.36.0.1036
Hardware management	7.0060.1523
NetQ	5.2.1
FRR	10.0.3
NVUE Core	1.14.0.81
UMF	91.1.14.14
BMC	88.0060.2408
BMC ERoT	02.00.0023.0000_n05

New Features

- New `nv show platform cpo` and `nv show platform laser-source` commands display Co-Packaged Optics (CPO) module and External Laser Source (ELS) status, including details for each channel and each laser, on the CPO-based switch. Refer to Show CPO and Laser-Source Commands.
- CPO module, optical engine, and ELS metrics are now available over OTEL and gNMI telemetry export, using the existing `transceiver-info platform-stats` class and a new `laser-source-info platform-stats` class.

Unsupported Features

The Spectrum-6 engineering sample does not support:

- > Optimized image upgrade
- > Package upgrade
- > ISSU upgrade
- > Fast reboot
- > Precision Time Protocol (PTP)

Network Port Connectivity Matrix

Supported Speeds	Connectivity	Cable PN	Cable Type and Description	Lengths (m)
800G_R4 to 200G_R1	SN6200-LD ↔ SN6810-LD/ ConnectX-9	980-9IAU1-00X M00	DR4 Optic module, OSFP flat	Up to 100
800G_R4 to 200G_R1	SN6200-LD ↔ SN6810-LD	980-9IAM6-00X 001	ACC copper, OSFP flat	1, 2, 3

When connecting to ConnectX-9 using cable PN 980-9IAU1-00XM00, ensure that the adapter is running firmware version 82.49.10xx or later and apply the configuration setting `mlxconfig -d /dev/mst/mt4133_pciconf0 set MODULE_MAX_POWER_OVERRIDE_M0=0x96`.

Service Port Connectivity Matrix

Supported Speeds	Connectivity	Cable PN	Cable Type and Description	Lengths (m)
100G_R1	SN6810-LD	980-9I068-00NM00 (MMS1X00-NS400)	DR4 Optic module, QSFP112 flat	Up to 100

Verified Cables

Speed	Part Number	Description
1600GbE	980-9IAU1-00XM00	NVIDIA twin port transceiver, 2xDR4, 1600Gbps, OSFP, 2xMPO(APC), 1310nm SMF, up to 500m, RHS, FRO, Gen2, Ethernet Only
1600GbE	980-9IAM6-00X001	NVIDIA Active copper cable, 1600Gbps to 1600Gbps, OSFP, 1.1m, RHS to RHS
1600GbE	980-9IAM6-00X002	NVIDIA Active copper cable, 1600Gbps to 1600Gbps, OSFP, 2m, RHS to RHS
1600GbE	980-9IAM6-00X003	NVIDIA Active copper cable, 1600Gbps to 1600Gbps, OSFP, 3m, RHS to RHS

NVIDIA DSX Air

NVIDIA DSX Air only supports 256 ports (the physical switch supports 512 ports).

Install BMC

The Spectrum-6 switch requires you to download and install the Baseboard Management Controller (BMC) onto the switch.

BMC is a specialized microcontroller designed to deliver out-of-band remote monitoring and management for servers and switches. Operating independently from the main CPU and operating system, the BMC enables administrative control even when the switch is powered down or unresponsive. It streamlines server and switch management while enhancing network efficiency, reliability, and security through automation of critical tasks.

To download the BMC package, run the `nv action fetch platform firmware <firmware-name> <remote-url-to-package>` command:

```
cumulus@switch:~$ nv action fetch platform firmware BMC <remote-url-to-BMC-package>
Action executing ...
Firmware downloaded successfully. File:
/tmp/firmware_downloads/sw_bmc_arm_chameleon5_chameleon_bu_kernel-001.fwpkg. SHA256:
19439f4a6695473386a9bb5f2fe253126825c0336890cd0e7366c7a10bda97c0
Action succeeded
```

To install the BMC package, run the `nv action install platform firmware <firmware-name> files <downloaded-file-name>` command:

```
cumulus@switch:~$ nv action install platform firmware BMC files <downloaded-file-name>
The operation will install the firmware.
Type [y] to install the firmware.
Type [N] to abort.
Do you want to continue? [y/N] y
Action executing ...
Firmware installation completed for bmc. Version: unknown
Action succeeded
```

If you prefer not to reboot the switch after BMC installation completes, add the `skip-reboot` command option; for example, `nv action install platform firmware BMC files <downloaded-file-name> skip-reboot`.

Show BMC Commands

Use the following commands to show information about the installed BMC package and switch environment leakage.

Command	Description
<code>nv show platform inventory</code>	Shows the model, serial number, operational state, and type of each component on the switch.
<code>nv show platform inventory BMC</code>	Shows the BMC firmware model, serial number, operational state, and type.
<code>nv show platform firmware BMC</code>	Shows the BMC firmware part number and name.
<code>nv show platform firmware BMC files</code>	Shows the available BMC firmware files.
<code>nv show platform environment leakage</code> <code>nv show platform environment leakage -o json</code>	Shows the switch environment leakage (low-current signals).

The following example shows the BMC model, serial number, operational state, and type.

```
cumulus@switch:~$ nv show platform inventory BMC
                        operational
-----
state                ok
hardware-version      chameleon_bu_kernel_leak-002
model                 Baseboard Management Controller
serial               MT2538600R0B
type                  bmc
```

The following examples show the switch environment leakage:

```
cumulus@switch:~$ nv show platform environment leakage
Name      State
-----
leakage1   normal
leakage2   normal
```

```
cumulus@switch:~$ nv show platform environment leakage -o json
{
  "leakage1": {
    "state": "normal"
  },
  "leakage2": {
    "state": "normal"
  }
}
```

Show CPO and Laser-Source Commands

A CPO-based switch organizes transceiver and laser status around Co-Packaged Optics (CPO) modules and External Laser Sources (ELS) instead of individual front-panel ports. Use the following commands to show CPO module and ELS status.

Command	Description
<code>nv show platform cpo</code>	Lists the CPO modules on the switch.
<code>nv show platform cpo <cpo-id></code>	Shows status, firmware version, port mapping, and the full channel and optical-engine readout for one CPO module.
<code>nv show platform cpo <cpo-id> channel</code>	Shows details for each channel for one CPO module.
<code>nv show platform cpo <cpo-id> optical-engine</code>	Shows the optical engines for one CPO module.
<code>nv show platform cpo <cpo-id> threshold</code>	Shows the alarm thresholds for one CPO module.
<code>nv show platform laser-source</code>	Lists the ELSs on the switch.
<code>nv show platform laser-source <laser-source-id></code>	Shows status and the full laser-specific readout for one laser source.
<code>nv show platform laser-source <laser-source-id> laser</code>	Shows laser-specific detail only for one laser source.
<code>nv show platform laser-source <laser-source-id> threshold</code>	Shows the alarm thresholds for one laser source.
<code>nv show interface <interface-id> cpo</code>	Shows the CPO module information scoped to one interface.

The following example lists the CPO modules on the switch.

```
cumulus@switch:~$ nv show platform cpo
```

CPO	Status	Error-status	FW Version	Ports	Laser Source	Optical Engines
----	-----	-----	-----	-----	-----	-----
cpo1	inserted	fiber contaminated	42.40.15	swp1-swp4	els1	oe1, oe2
cpo2	...	...	...	...	...	...

The following example lists the ELSs on the switch.


```
cumulus@switch:~$ nv show platform laser-source
```

Laser-source	Identifier	Vendor name	Vendor PN	Vendor SN	Vendor revision	FW Version
els1	ELS	NVIDIA	MCP7Y60-H01A	MT2233VS02221	A4	42.40.15
els2	ELS	NVIDIA	MCP7Y60-H01B	MT2233VS02222	A4	42.40.15

The `nv show platform cpo <cpo-id>` and `nv show platform laser-source <laser-source-id>` commands return the full channel, optical-engine, and laser readout for the module. Use the channel, optical-engine, laser, or threshold child commands shown above to limit the output to one part of a module.

CPO module, optical engine, and ELS metrics are also available over OTEL and gNMI telemetry export. For OTEL, enable export with the existing `transceiver-info platform-stats class` for CPO module and optical-engine metrics, and the new `laser-source-info platform-stats class` for ELS and per-laser metrics, for example:

```
cumulus@switch:~$ nv set system telemetry platform-stats class transceiver-info state enabled
cumulus@switch:~$ nv set system telemetry platform-stats class laser-source-info state enabled
```

Refer to the [Cumulus Linux 5.18 User Guide](#) for the full open telemetry configuration.

Liquid Cooling Leakage Telemetry

Liquid-cooled NVIDIA switches contain leakage sensors that detect coolant leaks. These sensors are critical for data center safety as an undetected leak can cause equipment damage, fire risk, or unplanned downtime.

This feature adds leakage sensor data to the gNMI and OTEL telemetry pipeline, enabling remote monitoring systems, such as Prometheus, Grafana, and gNMI collectors, to subscribe to real-time leakage status and trigger automated alerts.

Open Issues

ID	Description

Legal Notices and 3rd Party Licenses

Product	Version	Legal Notices and 3rd Party Licenses
SDK	4.10.0290	> 3rd Party Unify Notice > 3rd Party Notice > License
MFT	4.35.0-2064	> 3rd Party Notice—Linux > License—Linux

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